Inner imagery when listening to music

Introduction

Music listening may evoke mental imagery in certain listeners, though the prevalence and role of such imagery remain uncertain. Understanding whether and how mental imagery responses are influenced by music is therefore of interest. This study explores the occurrence of mental imagery during music listening and examines its potential relationship with emotional engagement and enjoyment. The paper's initial hypothesis is built on this examination.

Initial hypothesis

The majority of people listening to music are expected to generate mental imagery in response to musical stimuli.

Literature review

Mental imagery and music listening

Mental imagery is a common cognitive function that allows individuals to internally generate visual experiences in the absence of any external input. It is seen as an internal imitation of perception, and perception itself is subjective to each person, meaning mental imagery is not limited to only the visual domain. It is a multimodal in nature, encompassing auditory (such as mentally hearing music or voices), motor (imagining movement, gestures, or dance), and emotional components (the re-experiencing of moods or affective states associated with memories or environments) [1]. This multimodality suggests that mental imagery is closely linked to embodied cognition, where cognitive processes are grounded in bodily sensations, movement, and affective experience.

Music listening can, depending on the person, provide a rich basis for mental imagery [1, 2]. Differently than other auditory stimuli like speech, music unfolds over time and is structured through rhythm, harmony, melody, and timbre. When listening to music, people may experience auditory imagery in the form of mentally replaying sound, as well as visual imagery such as scenes, narratives, or environments following the internal emotional reaction to the music or associated by the music. Music is also frequently known to evoke motor imagery. This includes imagined movement, dancing, or gestures [3]. Previous research has indicated that musical imagery often occurs spontaneously during listening, even without explicit instructions [1, 3]. This suggests that music may naturally encourage internal simulation, imaginative engagement, and might make people think in more creative ways than without it. It is individual and, thus, subjective as to which degree listeners experiences mental imagery [2, 3]. It has been suggested that factors such as imagery ability, and familiarity with musical styles, or other creative interests could influence the vividness and frequency of imagery during listening to music, though this idea does not have enough proof, so this is not certain [1].

Mental imagery and emotion in music

The relationship between mental imagery and emotion is particularly relevant in the context of music. Imagery has been shown to intensify emotional responses by providing a narrative or sensory framework, which the listener will get while interpreting the music. Emotional engagement with music could be partly caused by the way the music sounds, and formed by the imagery the music may provoke [3]. From an embodied cognition perspective, mental imagery and mental imagery in music, can be understood as an internal activation of the person's perceptual systems, draws on stored perceptual knowledge and prior experience [1]. This activation could maybe enhance the vividness of mental imagery by reinforcing sensory imitation through a musical auditory input. Maybe listening to music could increase the intensity or vividness of mental imagery compared to a person's basis imagery ability, which is measured without musical stimuli. Taken together, existing research suggests that music listening has the potential to evoke and enhance mental imagery through its multimodal, embodied, and emotionally expressive qualities. Investigating this relationship may provide insight into how imagination contributes to emotional engagement and mental vividness during music listening [1, 2, 3].

This study will investigate whether listening to music influences the vividness of mental imagery compared to baseline imagery ability. Specifically, the study examines whether musical stimuli can enhance people's reported imagery vividness. This leads to the final hypothesis

Final hypothesis

People's mental imagery will become improved in response with musical stimuli.

Methods

Overview of test

To test the final hypothesis, drawing on the understanding that mental imagery is a subjective experience which creates a challenge on how to measure vividness results objectively, the questionnaire employed a two-part assessment:

The first part consisted of the *Vividness of Visual Imagery Questionnaire* (VVIQ) [4, 6], a well-established instrument used to assess individual differences in the vividness of visual mental imagery. The VVIQ [6] has been shown to reliably measure how clearly individuals can visualize images using their "*mind's eye*". Participants will respond to the VVIQ items using a 5-point Likert scale, ranging from "*no image at all*", *"only a knowledge of thinking about the object"* to "*perfectly clear and as vivid as normal vision"*. This initial section provides a quantitative baseline measure of each participant's general imagery vividness.

The second part of the assessment combined quantitative and qualitative measures. Participants will be asked to listen to three musical pieces "*Audio_1*," "*Audio_2*," and "*Audio_3*" and, after each piece, rate the vividness of their mental imagery using the same VVIQ scale. In addition, participants were given the option to describe in writing any imagery they experienced while listening, although providing a written response was not mandatory.

The questionnaire was divided into two parts to examine whether listening to music influenced the vividness of participants' mental imagery, compared to their baseline imagery ability measured by the VVIQ alone.

Music for the test

The musical stimuli were selected with the intention of conducting a pilot test. Three musical pieces representing different genres:

- Rock.
- Chill lo-fi.
- Upbeat pop.

They were used in order to evoke varying emotional and imaginative responses. The use of different genres was intended to simulate varied listening scenarios and examine whether different musical characteristics influenced mental imagery vividness. All musical pieces were sourced from "*freetouse.com*" [7].

Results

First part of the survey

The first part of the survey consisted of the Vividness of Visual Imagery questionnaire. These are the result collected from 10 different respondents based on the vividness responses explained in methods. The participants could select these responses to the different questions:

1. No image at all, you only know what you are thinking of the object.
2. Vague and dim.
3. Moderately clear and vivid.
4. Clear and reasonably vivid.
5. Perfectly clear and as vivid as normal vision.

It is put into a table with the numbers 1 to 5 as above. This is the response for each of the statements. It shows the totals for each statement:

No image at all, you only know what you are thinking of the object.	36
Vague and dim.	24
Moderately clear and vivid.	49
Clear and reasonably vivid.	44
Perfectly clear and as vivid as normal vision.	17

Second part of the survey

The second part is where the participants listened to music and gave their answer to how vivid their mental imagery was. This was the results:

No image at all, you only know what you are thinking of the object.	8
Vague and dim.	4
Moderately clear and vivid.	9

Clear and reasonably vivid.	8
Perfectly clear and as vivid as normal vision.	1

The participant's also got an optional question of what they imagined while listening to the audio-files. This table will go through these more qualitative responses:

Participant number	Audio_1 (Rock)	Audio_2 (Chill lofi)	Audio_3 (Upbeat pop)
1			
2	Concert, playing drums, party and bar	Tavern, folk music, love song	Advertisement for something, website,
3	Felt like being in a small room, playing instruments with some others.	Just pleasant music, didn't give me the feeling of being present in a setting.	Kinda weird, it gave me the feeling of moments from my past, but nothing concrete. I think it is because the first notes reminded me of a song from a long time ago.
4	I see my friends and myself on a rock concert jumping around and smiling. Nearly there's pogo. It's a dark room, some lighting that blinds me slightly	A Japanese tea time in a garden with girls in light-colour clothes.	Roblox 😳
5	A band playing music on a scene, like small concert thing. Guitar, drums and such.	A girl sitting in a window sill/ledge thingy with multiple pillows and a warm fuzzy blanket, while having a warm teacup in her hands with cat motives on it. She is looking out the window at the rain calmly pouring down.	Corporate video that is very basic and "trendy"
6	A recording studio	A trio of guitarists	It is summertime.

		playing.	Beachwalks, out in the park enjoying the flowers, a good cold drink, sunglasses and very light with the sun shining, hanging with friends.
7	A scene from an archetypal 2000's teenage movie - think high school musical - going from the hallways of a high school to the gymnasium/meeting hall.	walking through a medieval fantasy town, specifically through the marketplace of said town with booths and stands covering both sides of the street.	Windmills, people smiling in an office at a whiteboard.
8	A band practicing.	A log cabin	travel/airplane commercials
9	The club going crazy in a blurry mess of party and vibes. As well as a motorcycle ride down the highway	Sitting by the mediterarrian sea and enjoying a nice stroll through a small city by the coast, as the time is around 8 pm and the sun is slowly setting.	A YouTube tutorial, with notepad in place of a microphone.
10			I am watching a youtube tutorial where a notepad appears on screen explaining me how to build a house in minecraft
All in all	Concerts were mentioned three times, band practicing, clubs, and recording studio. These were the themes in the participant's imaginations.	A lot of the imagery was very chill. Strolling and enjoying the scenery, drinking tea or being comfortable under blankets, being in a garden, and a log cabin, or girl in colorful clothes. It was relaxing imagery.	The upbeat music was either seen as something corporate or from commercials, or music from a YouTube tutorial where someone uses a notepad instead of talking. Other than music used in media, other people thought about something

			nostalgic or walks in the beach or a park, hanging with people.

The difference

As the results from the first survey outweighs the second survey answer by four, due to there being more questions in the first survey, the results were divided by four before the difference between the results could be shown. This was done, to normalize for differing numbers of the questionnaire items. This is how it was done:

Total from first test		
1	36	
2	24	
3	49	
4	44	
5	17	
Total from first test Divided		
1	9	
2	6	
3	12,25	
4	11	
5	4,25	

This shows the results in percent:

%
21,18
14,12
28,82
25,88
10,00

This was the result from the second part survey with music that the total from the first test which had been divided was compared to:

Total from second test	
1	8
2	4
3	9
4	8
5	1

This shows the results in percent:

%
26,67
13,33
30,00
26,67
3,33

This is the difference between the two results:

Diffrance	
1	1
2	2
3	3,25
4	3
5	3,25

Discussion

The results show that the correlation between the first part of the survey or the baseline responses, which had been divided, and the second part of the survey responses didn't have the biggest difference numerically, though it indicates that the first beats the second by a few. Even though it appears this way, it is that the first part of the survey has four times more questions, and has 40 responses total. The music table has 30 responses, so it would be focusing on the wrong thing to ask which condition had more answers. Instead, it was looked at how vivid imagery was for each response. The choice to normalize the answers from the baseline of the test, was necessary for looking into this difference.

In general, the music condition of the test has fewer "no image at all" than the first part of the test. As if music helped guide people's mental image. After normalization of totals, imagery vividness during music listening was comparable to, and in some categories slightly higher than, baseline imagery vividness. Although overall vividness of mental images did not increase dramatically, the reduction in reports of absent imagery during music listening, suggests that musical stimuli may facilitate the generation of mental imagery in some listeners.

Conclusions

The study investigated mental imagery before and during music listening and examined whether musical stimuli influenced the vividness of the imagery compared to the baseline imagery results. The results showed that participants frequently experienced mental imagery while listening to music, and that different musical genres helped built the mental images narratives.

While imagery vividness during music listening did not increase substantially compared to the baseline imagery ability, there were fewer reports of not having any images, when the music was present. This suggest that music may support mental imagery rather than significantly enhancing vividness. Given that this was a pilot test which had a small sample size, the findings did not give substantial proof of the final hypothesis, and should be viewed cautiously. Even though that is the case, the study provides preliminary insights into the relationship between music listening and mental imagery and can highlight the need for further research using larger and more controlled samples.

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